BRAC IT CodeSprint 2026
Project BRD

Purpose: Explain your idea clearly enough that a registered team, who did not submit it but can read it, understand it, and decide to build it.
Recommended length: 2–4 pages. Short, clear bullet points beat long paragraphs to write for a team meeting this idea for the first time.

1. Project Name : **BRAC স্বাস্থ্যকর্মী প্রশিক্ষণ - AI সিমুলেটর**

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2. Project Owner : **Primarily - BRAC IT**

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3. Problem Statement
What problem are you trying to solve?

Community health workers often finish training with theoretical knowledge but little practical experience talking to real patients. This creates a gap between classroom learning and fieldwork, where workers must handle sensitive health conversations, patient concerns, and unexpected questions with confidence. The project solves this by providing a safe, AI-powered simulator where workers can repeatedly practice realistic patient conversations in Bengali and receive feedback before meeting actual patients.

4. Proposed Solution
Explain your solution in detail

The proposed solution is BRAC Health Mentor Sim, an AI-powered conversational training platform that lets community health workers practice patient interactions in a safe, simulated environment.

Workers log into the app and choose from realistic training scenarios — for example, a mother concerned about her child's fever, a patient unsure about taking medication, or a family asking about family planning. Each scenario is powered by a large language model acting as a virtual patient, responding naturally in Bengali with emotions, concerns, and questions just like a real person.

During a session, the worker types or speaks their response, and the AI patient reacts in real time. The conversation continues until the worker resolves the situation or reaches a natural stopping point. Afterward, the system gives structured feedback: what the worker did well, where they missed a key concern, whether they used clear and respectful language, and how they could improve. Workers can repeat scenarios as many times as they want, try different approaches, and build confidence before facing similar conversations in the field.

The platform also tracks progress over time, so trainers and supervisors can see which workers need more support and which skills are improving across the team.
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5. Target Users
Who will use / benefit from it?

The solution has three main user groups:

Primary users
- BRAC community health workers — frontline workers who visit homes, provide basic health services, and counsel families. They use the simulator to practice patient conversations in Bengali before going into the field.
- Trainers and supervisors — they assign scenarios, review worker performance, and identify who needs extra coaching.

Secondary users
- Program managers — they track training progress across regions and measure whether communication skills are improving over time.

Beneficiaries
- Patients and communities — they receive clearer, more empathetic, and more effective health guidance from better-prepared workers.
- BRAC — it gets scalable, repeatable training that reduces the cost and risk of relying only on in-person role-play.

6. Key Features
What should the application be able to do? (a short list is fine)

The application should be able to do the following:

Core training features
– Run AI-powered patient conversations in Bengali, where a language model acts as a realistic virtual patient.
– Offer a library of training scenarios covering common community health topics: maternal health, child illness, nutrition, family planning, medication adherence, and infectious disease awareness.
– Accept typed or spoken responses from health workers and react naturally in real time.
– Provide structured feedback after each session on communication clarity, empathy, missing concerns, and clinical accuracy.

Progress and coaching
– Track each worker's practice history, improvement trends, and weakest skill areas.
– Let trainers assign specific scenarios and review session transcripts and scores.
– Highlight workers who need extra support and celebrate consistent improvement.

Platform features
– Support role-based access for workers, trainers, supervisors, and program managers.
– Work on low-bandwidth devices and basic smartphones, since many health workers use entry-level phones in rural areas.
– Keep a lightweight, simple interface that feels familiar and easy to use without extensive tech training.

7. AI Component
Where exactly will AI be used?

AI will be used in three specific places inside the application:

1. Virtual patient conversations
The core AI role is to act as the simulated patient. A language model plays the patient in each scenario — responding in Bengali, showing emotion, asking follow-up questions, and reacting naturally to whatever the health worker says. This makes each practice session feel like a real home visit.

2. Real-time feedback and scoring
After a session ends, AI evaluates the full conversation transcript. It scores the worker on clarity, empathy, completeness, and whether they addressed the patient's actual concern. It then produces structured, actionable feedback the worker can read and learn from.

3. Scenario adaptation and coaching insights
AI can adjust scenario difficulty based on the worker's progress and suggest the next scenario to practice. For supervisors, AI can summarize patterns across many sessions — for example, flagging that several workers struggle with medication-adherence conversations — so trainers know where to focus.

Optional AI layers include speech-to-text for workers who prefer speaking over typing, and text-to-speech so the virtual patient can talk back, making the simulator feel more natural on basic smartphones.

8. Expected Outcome / Impact
What improvement will the solution create?

The solution is expected to create measurable improvements at three levels:

For health workers
– Higher confidence when speaking with real patients, because they have already practiced similar conversations many times.
– Faster skill development through immediate, specific feedback instead of waiting for rare in-person role-play sessions.
– A safe space to make mistakes and try different approaches without risking patient trust or health.

For training programs
– Scalable, repeatable training that does not depend on finding human actors or scheduling live sessions.
– Clear data on who needs help and in which skill areas, so supervisors can coach more efficiently.
– Lower training costs per worker while maintaining consistent quality across regions.

For patients and communities

– Better health counseling, because workers communicate more clearly and empathetically.
– Improved health outcomes over time, such as higher medication adherence, better maternal and child health awareness, and more timely referrals.
– Greater trust in frontline health workers, since patients feel heard and understood.

9. Data / Resources Required
What information, systems, APIs, or datasets might be needed?

The project will likely need the following resources:

AI and APIs
– A language model API for the virtual patient and feedback engine. Lovable AI Gateway can handle this through the app.
– Optional speech-to-text and text-to-speech APIs if workers use voice instead of typing.
– A gateway key and credits for AI usage.

Backend and data
– A database to store users, training scenarios, conversation sessions, feedback scores, and progress history. Lovable Cloud can provide this with built-in authentication.
– Authentication for workers, trainers, supervisors, and program managers with role-based access.

Content and domain knowledge
– Realistic patient scenarios developed with BRAC trainers, covering the actual health issues workers face in the field.
– Conversation examples and rubrics from experienced health workers or supervisors, so the AI gives feedback that matches BRAC standards.
– Medical accuracy review to ensure the simulator does not teach incorrect advice.

Operational data
– Worker profiles, assigned regions, training cohorts, and supervisor relationships.
– Analytics to track engagement, skill improvement, and scenario completion rates.

Device and access requirements
– A lightweight web app that works on entry-level smartphones and unstable internet connections, since many frontline workers operate in rural areas.

10. Success Criteria
How will we know the solution works?

We will know the solution works by tracking both usage and outcome metrics:

Adoption and engagement
– At least 70% of enrolled health workers complete at least one practice session per week.
– Average session length shows workers are engaged, not quitting early.
– Trainers and supervisors actively assign scenarios and review feedback.

Learning improvement
– Workers' feedback scores improve over repeated sessions in the same scenario type.
– Pre- and post-assessments show measurable gains in communication skills and clinical accuracy.
– Workers report higher confidence before home visits after using the simulator for a few weeks.

Real-world impact
– Supervisors observe better patient interactions during field visits.
– Patient satisfaction scores or trust ratings for health workers increase.
– Health outcomes move in the right direction — for example, more timely referrals, higher medication adherence, or better follow-up rates.

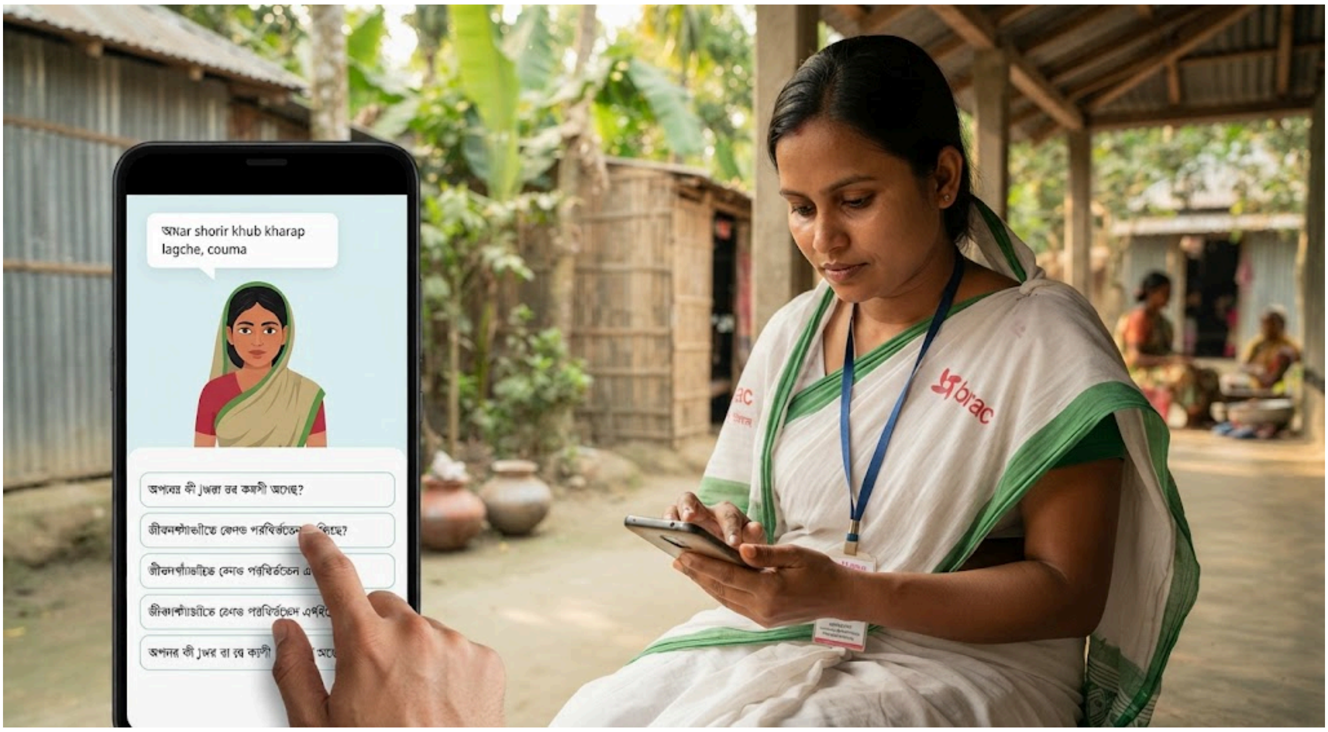
Operational efficiency
– Training time per worker decreases while quality stays the same or improves.
– Supervisors spend less time on basic role-play and more time coaching workers who actually need support.

Qualitative signals
– Workers say the simulator feels realistic and useful.
– Trainers confirm the feedback matches what they would say in person.
– The app remains easy to use on the devices workers already have.

Why Should You Build This?

Why Should You Build This?

In 3–5 lines, make the case for your idea. This is what teams will read side by side across all 10 ideas. Be specific about the impact and why it's worth their time.



The opportunity is clear BRAC trains thousands of community health workers, but classroom time is limited and real patients are not practice dummies. Workers need safe, repeatable conversation practice before they knock on doors. An AI simulator gives them exactly that, at scale, without pulling patients or trainers into every session.

It is technically feasible and cost-effective Large language models can now hold realistic conversations in Bengali, and the app can run on basic smartphones. The infrastructure — authentication, database, and AI — can be handled through Lovable Cloud and Lovable AI, so the team can focus on the training content rather than building backend systems from scratch.

The impact is measurable Better-prepared health workers lead to clearer counseling, stronger patient trust, and ultimately better health outcomes in underserved communities. The app also produces data that helps supervisors coach smarter and programs allocate training resources where they matter most.

It aligns with BRAC's mission The project extends BRAC's existing frontline health work by giving workers a modern, practical tool to improve the human side of care — communication, empathy, and confidence.

Submitting: Save this as a PDF and upload it through the BRD Submission Form, along with your basic details (name, employee PIN, project name). Supporting materials, a wireframe, a sample, or reference links are optional but welcome.